

The empirical recurrence for OEIS A236060, recovered from its tail

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Abstract

OEIS A236060 records “Empirical recurrence of order 94” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1616$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 102$. Its last coefficient is $c_{94} = -519356416 \neq 0$, so no further tail satisfies anything shorter; any order-94 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A236060 is “Number of $(n+1) \times (3+1) \times 0..2$ arrays colored with the difference of the upper median and the minimum in each 2×2 subblock.”. It has offset 1 and begins

1616, 8506, 45987, 221292, 1178874, 6003004, 32549159, 160724066, ...

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:03:41, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1616$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1616$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 94: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 94.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3232$ exact terms from $n = 8$ on, it returns order 94 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	0	c_{25}	−10533527924	c_{49}	974787116470920	c_{73}	261921639599556
c_2	0	c_{26}	37616029484	c_{50}	10787053451733722	c_{74}	−444928371683606
c_3	0	c_{27}	−3235107994	c_{51}	1635078352075522	c_{75}	138915275607280
c_4	670	c_{28}	−821124706403	c_{52}	19515301092162490	c_{76}	−160326836761924
c_5	4	c_{29}	146269806628	c_{53}	310154930817594	c_{77}	72321842043080
c_6	5498	c_{30}	−4552988184384	c_{54}	25514633993204913	c_{78}	−71729262403964
c_7	148	c_{31}	361595542390	c_{55}	−4558695507747418	c_{79}	−7584606819440
c_8	−81785	c_{32}	−6833549376759	c_{56}	26402544179515508	c_{80}	−17480127330072
c_9	1266	c_{33}	792657746524	c_{57}	−7571990267089218	c_{81}	−6907164278384
c_{10}	−836232	c_{34}	26984294532329	c_{58}	19255717262821841	c_{82}	6983694210408
c_{11}	−2798	c_{35}	−1789580760474	c_{59}	−7400733448694610	c_{83}	−2998084947552
c_{12}	3479313	c_{36}	113070277365055	c_{60}	14477764525918391	c_{84}	5990705178064
c_{13}	−221342	c_{37}	−15742149147544	c_{61}	−6559880474494338	c_{85}	−3907624570048
c_{14}	49511058	c_{38}	111449920976075	c_{62}	12753306702934939	c_{86}	1780685441760
c_{15}	−1608754	c_{39}	−12629089954460	c_{63}	−3972572262177680	c_{87}	−917377124992
c_{16}	10667878	c_{40}	−223544534401691	c_{64}	6303347350300022	c_{88}	398270335040
c_{17}	9746666	c_{41}	646835922432	c_{65}	−1749768463046508	c_{89}	−163314237440
c_{18}	−1514082434	c_{42}	−878383486894378	c_{66}	1232756251732448	c_{90}	39673188864
c_{19}	97698494	c_{43}	−12179240562224	c_{67}	−944322717023844	c_{91}	−74098628608
c_{20}	−4901221534	c_{44}	−1952472896231725	c_{68}	846390775377926	c_{92}	−2686113280
c_{21}	851108	c_{45}	263740484879848	c_{69}	159082745611180	c_{93}	−12124295168
c_{22}	23402811478	c_{46}	−2856847896602143	c_{70}	255131944863342	c_{94}	−519356416
c_{23}	−1591673874	c_{47}	701965212107346	c_{71}	611957029372644		
c_{24}	116493563299	c_{48}	995328082146859	c_{72}	−602532143256224		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 102$, and it is the recurrence OEIS A236060 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{94} - c_1x^{94-1} - \dots - c_{94}$. Its constant term is $-c_{94} = 519356416$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 94 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 94, so $\deg q = 94$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 94, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 94, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -519356416 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-94 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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